

cppulse

C++ Codebase Health Report

/tmp/poco

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1. Executive Summary

97.8Overall Health Score
Good — codebase is in healthy condition.

25821

TOTAL FINDINGS

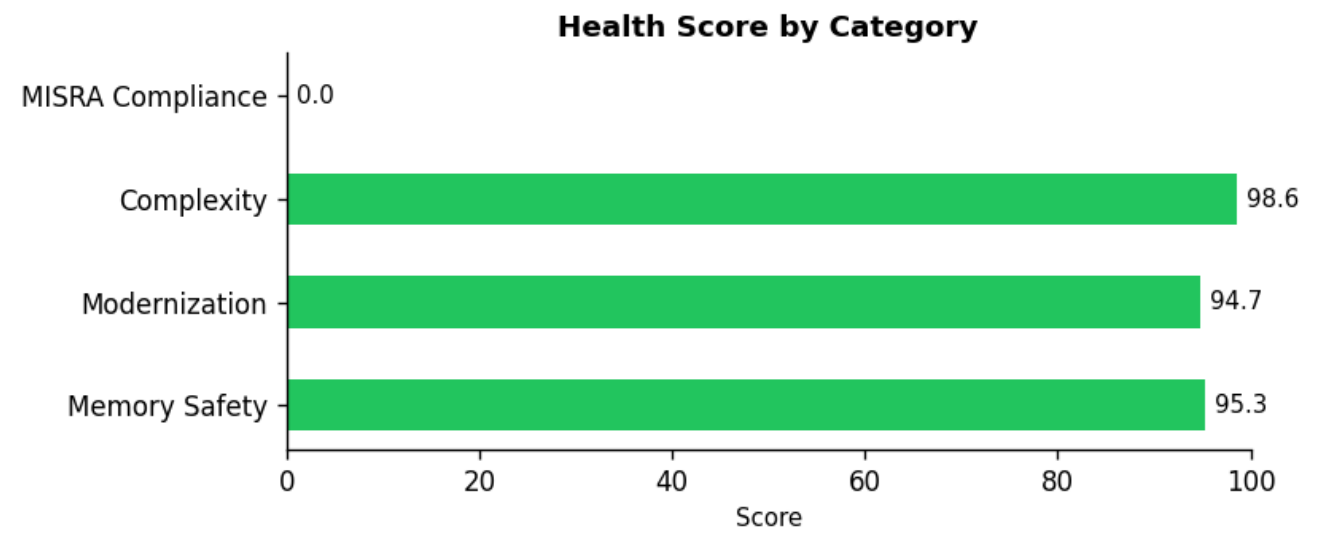
545

ERRORS

16929

WARNINGS

2. Health Score Breakdown



Category	Score (/ 100)	Status
Memory Safety	95.3	GOOD
Modernization	94.7	GOOD
Complexity	98.6	GOOD
MISRA Compliance	0.0	POOR

3. Hotspot Map — Top 20 Files

Hotspot score = change frequency × complexity × debt density. High-score files deliver the greatest ROI when refactored.

#	File	Hotspot Score	Change Freq.	Findings
1	ActiveRecord/ Compiler/src/ CodeGenerator.cpp	0.00	0	7
2	ActiveRecord/ Compiler/src/ Compiler.cpp	0.00	0	5
3	ActiveRecord/ Compiler/src/ HeaderGenerator.cpp	0.00	0	6
4	ActiveRecord/ Compiler/src/ ImplGenerator.cpp	0.00	0	11
5	ActiveRecord/ Compiler/src/ Parser.cpp	0.00	0	5
6	ActiveRecord/ include/Poco/ ActiveRecord/ Query.h	0.00	0	5
7	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	0.00	0	29
8	ActiveRecord/ testsuite/src/ ActiveRecordTestSuite.cpp	0.00	0	1
9	ActiveRecord/ testsuite/src/ WinDriver.cpp	0.00	0	1
10	ApacheConnector/ include/ ApacheApplication.h	0.00	0	3
11	ApacheConnector/ include/ ApacheConnector.h	0.00	0	1
12	ApacheConnector/ include/ ApacheRequestHandlerFactory.h	0.00	0	3

#	File	Hotspot Score	Change Freq.	Findings
13	ApacheConnector/ include/ ApacheServerReques t.h	0.00	0	2
14	ApacheConnector/ include/ ApacheServerRespon se.h	0.00	0	2
15	ApacheConnector/ include/ ApacheStream.h	0.00	0	2
16	ApacheConnector/ samples/ FormServer/src/ FormServer.cpp	0.00	0	9
17	ApacheConnector/ samples/ TimeServer/src/ TimeServer.cpp	0.00	0	3
18	ApacheConnector/ src/ ApacheConnector.cp p	0.00	0	6
19	Benchmark/src/ BenchmarkApp.cpp	0.00	0	18
20	Benchmark/src/ LoggerBench.cpp	0.00	0	21

4. Detection Findings

Complexity (7 rule violations)

Rule	Severity	Description	Files Affected	Occurrences
CPP-CX-001	ERROR	Function 'bareboneODBCTest' has cyclomatic complexity of 71 (threshold: warning>15, error>25) <i>Decompose into smaller functions to reduce complexity</i>	Data/ODBC/testsuite/src/SQLExecutor.cpp	1
CPP-CX-002	WARNING	Function 'writeInsert' is 83 lines long (threshold: warning>80, error>150) <i>Decompose into smaller helper functions</i>	ActiveRecord/Compiler/src/ImplGenerator.cpp	1
CPP-CX-003	WARNING	Function 'printTimeSummary' has 6 parameters (threshold: warning>5, error>8) <i>Introduce a parameter struct or builder pattern to reduce parameter count</i>	Benchmark/src/BenchmarkApp.cpp	1
CPP-CX-002	WARNING	Function 'printTimeSummary' is 110 lines long (threshold: warning>80, error>150) <i>Decompose into smaller helper functions</i>	Benchmark/src/BenchmarkApp.cpp	1
CPP-CX-002	WARNING	Function 'main' is 94 lines long (threshold: warning>80, error>150) <i>Decompose into smaller helper functions</i>	Benchmark/src/BenchmarkApp.cpp	1

Rule	Severity	Description	Files Affected	Occurrences
CPP-CX-002	WARNING	Function 'testRSAEVPPKey' is 100 lines long (threshold: warning>80, error>150) <i>Decompose into smaller helper functions</i>	Crypto/testsuite/src/EVPTest.cpp	1
CPP-CX-002	WARNING	Function 'testECEVPPKey' is 93 lines long (threshold: warning>80, error>150) <i>Decompose into smaller helper functions</i>	Crypto/testsuite/src/EVPTest.cpp	1

Memory Safety (3 rule violations)

Rule	Severity	Description	Files Affected	Occurrences
CPP-MEM-001	WARNING	Raw 'new' expression detected; prefer std::make_unique or std::make_shared <i>Replace 'new T(...)' with 'std::make_unique<T>(...)'</i>	ActiveRecord/include/Poco/ActiveRecord/Query.h, ApacheConnector/samples/FormServer/src/FormServer.cpp, ApacheConnector/samples/TimeServer/src/TimeServer.cpp +2 more	8
CPP-MEM-002	WARNING	Explicit 'delete' detected; prefer smart pointers for automatic lifetime management <i>Use std::unique_ptr or std::shared_ptr to avoid manual delete</i>	Crypto/src/EVPCipherImpl.cpp, Crypto/src/RSACipherImpl.cpp, Crypto/testsuite/src/EVPTest.cpp	7

Rule	Severity	Description	Files Affected	Occurrences
CPP-MEM-003	WARNING	C-style array parameter 'anOIDArray' decays to a raw pointer with no bounds information <i>Replace C-style array parameter with <code>std::span<T></code> or <code>std::vector<T></code></i>	Data/PostgreSQL/testsuite/src/PostgreSQLTest.cpp	1

Modernization (25 rule violations)

Rule	Severity	Description	Files Affected	Occurrences
CPP-MOD-004	INFO	String literal contains 3 escape sequences; consider a raw string literal for readability <i>Replace escaped string with <code>R"(...)"</code> raw string literal</i>	ActiveRecord/Compiler/src/CodeGenerator.cpp, ActiveRecord/Compiler/src/HeaderGenerator.cpp, ActiveRecord/Compiler/src/ImplGenerator.cpp +2 more	15
CPP-MOD-008	WARNING	Unscoped enum 'LogLevel' pollutes the enclosing namespace and allows implicit integer conversion <i>Replace 'enum LogLevel' with 'enum class LogLevel'</i>	ApacheConnector/include/ApacheApplication.h, ApacheConnector/include/ApacheConnector.h, ApacheConnector/include/ApacheRequestHandlerFactory.h +3 more	6

Rule	Severity	Description	Files Affected	Occurrences
CPP-MOD-008	WARNING	Unscoped enum '(unnamed enum at / tmp/poco/ ApacheConnector/ include/ ApacheStream.h: 36:2)' pollutes the enclosing namespace and allows implicit integer conversion <i>Replace 'enum (unnamed enum at / tmp/poco/ ApacheConnector/ include/ ApacheStream.h: 36:2)' with 'enum class (unnamed enum at / tmp/poco/ ApacheConnector/ include/ ApacheStream.h: 36:2)'</i>	ApacheConnector/ include/ ApacheApplication. h, ApacheConnector/ include/ ApacheRequestHandl erFactory.h, ApacheConnector/ include/ ApacheServerReques t.h +2 more	5
CPP-MOD-006	INFO	Index-based for loop may be replaceable with a range-based for loop <i>Consider 'for (const auto& elem : container)' instead</i>	Benchmark/src/ LoggerBench.cpp, Crypto/testsuite/ src/ DigestEngineTest.c pp	4
CPP-MOD-008	WARNING	Unscoped enum '(unnamed enum at / tmp/poco/Crypto/ testsuite/src/ CryptoTest.h:24:2)' pollutes the enclosing namespace and allows implicit integer conversion <i>Replace 'enum (unnamed enum at / tmp/poco/Crypto/ testsuite/src/ CryptoTest.h:24:2)' with 'enum class (unnamed enum at / tmp/poco/Crypto/ testsuite/src/ CryptoTest.h:24:2)'</i>	Crypto/testsuite/ src/ CryptoTest.cpp, Crypto/testsuite/ src/CryptoTest.h, Crypto/testsuite/ src/ CryptoTestSuite.cp p	3

Rule	Severity	Description	Files Affected	Occurrences
CPP-MOD-009	INFO	typedef 'RequestHandlerFactories' should use a C++11 type alias instead <i>Replace 'typedef int RequestHandlerFactories' with 'using RequestHandlerFactories = int'</i>	ApacheConnector/include/ApacheApplication.h, ApacheConnector/include/ApacheRequestHandlerFactory.h	2
CPP-MOD-001	WARNING	C-style cast bypasses C++ type-safety checks <i>Replace with static_cast<T>, dynamic_cast<T>, or reinterpret_cast<T></i>	Crypto/src/X509Certificate.cpp	1
CPP-MOD-002	WARNING	Deprecated C header <stdlib.h> included; prefer the C++ equivalent <i>Replace <stdlib.h> with <cstdlib></i>	Data/SQLParser/src/SQLParser.cpp	1
CPP-MOD-003	WARNING	Virtual method 'what' overrides a base class method but is missing the 'override' specifier <i>Add 'override' to the method declaration</i>	Data/SQLParser/test/auto_query_file_test.cpp	1
CPP-MOD-005	INFO	Verbose iterator type 'const std::filesystem::directory_iterator' can be simplified with 'auto' <i>Replace the explicit iterator type with 'auto'</i>	Data/samples/DBLogger/src/SQLLogInserter.cpp	1
CPP-MOD-007	WARNING	NULL macro used as null pointer constant; prefer nullptr <i>Replace NULL with nullptr</i>	Data/SQLParser/src/parser/flex_lexer.cpp	1

Rule	Severity	Description	Files Affected	Occurrences
CPP-MOD-004	INFO	String literal contains 5 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	ActiveRecord/ Compiler/src/ Compiler.cpp	1
CPP-MOD-004	INFO	String literal contains 65 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	ApacheConnector/ samples/ FormServer/src/ FormServer.cpp	1
CPP-MOD-004	INFO	String literal contains 4 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	ApacheConnector/ samples/ TimeServer/src/ TimeServer.cpp	1
CPP-MOD-004	INFO	String literal contains 8 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	Benchmark/src/ BenchmarkApp.cpp	1
CPP-MOD-004	INFO	String literal contains 21 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	CppParser/ testsuite/src/ CppParserTest.cpp	1
CPP-MOD-004	INFO	String literal contains 17 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	CppParser/ testsuite/src/ CppParserTest.cpp	1

Rule	Severity	Description	Files Affected	Occurrences
CPP-MOD-009	INFO	typedef 'EVP_CIPHER' should use a C++11 type alias instead <i>Replace 'typedef struct evp_cipher_st EVP_CIPHER' with 'using EVP_CIPHER = struct evp_cipher_st'</i>	Crypto/include/Poco/Crypto/CipherKeyImpl.h	1
CPP-MOD-008	WARNING	Unscoped enum 'RSAPaddingMode' pollutes the enclosing namespace and allows implicit integer conversion <i>Replace 'enum RSAPaddingMode' with 'enum class RSAPaddingMode'</i>	Crypto/include/Poco/Crypto/Crypto.h	1
CPP-MOD-009	INFO	typedef 'BIGNUM' should use a C++11 type alias instead <i>Replace 'typedef struct bignum_st BIGNUM' with 'using BIGNUM = struct bignum_st'</i>	Crypto/include/Poco/Crypto/RSAPKeyImpl.h	1
CPP-MOD-009	INFO	typedef 'RSA' should use a C++11 type alias instead <i>Replace 'typedef struct rsa_st RSA' with 'using RSA = struct rsa_st'</i>	Crypto/include/Poco/Crypto/RSAPKeyImpl.h	1
CPP-MOD-008	WARNING	Unscoped enum 'Direction' pollutes the enclosing namespace and allows implicit integer conversion <i>Replace 'enum Direction' with 'enum class Direction'</i>	Crypto/src/CipherImpl.cpp	1
CPP-MOD-004	INFO	String literal contains 25 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	Crypto/testsuite/src/CryptoTest.cpp	1

Rule	Severity	Description	Files Affected	Occurrences
CPP-MOD-004	INFO	String literal contains 19 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	Crypto/testsuite/src/CryptoTest.cpp	1
CPP-MOD-004	INFO	String literal contains 96 escape sequences; consider a raw string literal for readability <i>Replace escaped string with R"(...)" raw string literal</i>	Crypto/testsuite/src/EVPTest.cpp	1

Misra (30 rule violations)

Rule	Severity	Description	Files Affected	Occurrences
MISRA-007	WARNING	Variable 'result' of type 'std::string' declared without an initializer <i>Initialize 'result' at the point of declaration</i>	ActiveRecord/Compiler/src/Parser.cpp, CppParser/src/Utility.cpp, Crypto/testsuite/src/CryptoTest.cpp	15
MISRA-007	WARNING	Variable 'msg' of type 'int' declared without an initializer <i>Initialize 'msg' at the point of declaration</i>	Benchmark/src/LoggerBench.cpp, Benchmark/src/NotificationQueueBench.cpp, Benchmark/src/PatternFormatterBench.cpp +3 more	15
MISRA-007	WARNING	Variable 'out' of type 'std::string' declared without an initializer <i>Initialize 'out' at the point of declaration</i>	Crypto/testsuite/src/CryptoTest.cpp	13
MISRA-007	WARNING	Variable 'result' of type 'int' declared without an initializer <i>Initialize 'result' at the point of declaration</i>	ActiveRecord/Compiler/src/CodeGenerator.cpp, ActiveRecord/include/Poco/ActiveRecord/Query.h, Benchmark/src/PatternFormatterBench.cpp	12

Rule	Severity	Description	Files Affected	Occurrences
MISRA-007	WARNING	Variable 'st' of type 'int' declared without an initializer <i>Initialize 'st' at the point of declaration</i>	CppParser/ testsuite/src/ CppParserTest.cpp	12
MISRA-007	WARNING	Variable 'pContext' of type 'int' declared without an initializer <i>Initialize 'pContext' at the point of declaration</i>	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	11
MISRA-007	WARNING	Variable 'pSuite' of type 'int *' declared without an initializer <i>Initialize 'pSuite' at the point of declaration</i>	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp, ActiveRecord/ testsuite/src/ ActiveRecordTestSuite.cpp, CppParser/ testsuite/src/ AttributesParserTest.cpp +7 more	10
MISRA-007	WARNING	Variable 'fmt' of type 'int' declared without an initializer <i>Initialize 'fmt' at the point of declaration</i>	Benchmark/src/ LoggerBench.cpp, Crypto/src/ EVPCipherImpl.cpp	10
MISRA-007	WARNING	Variable 'formatter' of type 'int' declared without an initializer <i>Initialize 'formatter' at the point of declaration</i>	Benchmark/src/ PatternFormatterBench.cpp	10
MISRA-007	WARNING	Variable 'key' of type 'int' declared without an initializer <i>Initialize 'key' at the point of declaration</i>	Crypto/samples/ genrsakey/src/ genrsakey.cpp, Crypto/testsuite/ src/ CryptoTest.cpp, Crypto/testsuite/ src/EVPTest.cpp	10
MISRA-007	WARNING	Variable 'parser' of type 'int' declared without an initializer <i>Initialize 'parser' at the point of declaration</i>	ActiveRecord/ Compiler/src/ Parser.cpp, CppParser/ testsuite/src/ AttributesParserTest.cpp, CppParser/ testsuite/src/ CppParserTest.cpp	9

Rule	Severity	Description	Files Affected	Occurrences
MISRA-007	WARNING	Variable 'pCipher' of type 'int' declared without an initializer <i>Initialize 'pCipher' at the point of declaration</i>	Crypto/testsuite/src/CryptoTest.cpp	8
MISRA-007	WARNING	Variable 'attrs' of type 'int' declared without an initializer <i>Initialize 'attrs' at the point of declaration</i>	CppParser/testsuite/src/AttributesParserTest.cpp	7
MISRA-007	WARNING	Variable 'result' of type 'auto' declared without an initializer <i>Initialize 'result' at the point of declaration</i>	ActiveRecord/testsuite/src/ActiveRecordTest.cpp	6
MISRA-007	WARNING	Variable 'source' of type 'int' declared without an initializer <i>Initialize 'source' at the point of declaration</i>	Benchmark/src/LoggerBench.cpp	5
MISRA-007	WARNING	Variable 'err' of type 'unsigned long' declared without an initializer <i>Initialize 'err' at the point of declaration</i>	Crypto/include/Poco/Crypto/Crypto.h, Crypto/src/CipherImpl.cpp, Crypto/src/CipherKeyImpl.cpp +2 more	5
MISRA-007	WARNING	Variable 'key2' of type 'int' declared without an initializer <i>Initialize 'key2' at the point of declaration</i>	Crypto/testsuite/src/EVPTest.cpp	5
MISRA-007	WARNING	Variable 'keyNE' of type 'int' declared without an initializer <i>Initialize 'keyNE' at the point of declaration</i>	Crypto/testsuite/src/EVPTest.cpp	5
MISRA-007	WARNING	Variable 'self' of type 'int' declared without an initializer <i>Initialize 'self' at the point of declaration</i>	ActiveRecord/Compiler/src/Compiler.cpp, Benchmark/src/BenchmarkApp.cpp, Crypto/samples/genrsakey/src/genrsakey.cpp	4

Rule	Severity	Description	Files Affected	Occurrences
MISRA-007	WARNING	Variable 'text' of type 'int' declared without an initializer <i>Initialize 'text' at the point of declaration</i>	Benchmark/src/LoggerBench.cpp	4
MISRA-007	WARNING	Variable 'queue' of type 'int' declared without an initializer <i>Initialize 'queue' at the point of declaration</i>	Benchmark/src/NotificationQueueBench.cpp	4
MISRA-007	WARNING	Variable 'pSym' of type 'int *' declared without an initializer <i>Initialize 'pSym' at the point of declaration</i>	CppParser/testsuite/src/CppParserTest.cpp	4
MISRA-007	WARNING	Variable 'pNSA' of type 'int *' declared without an initializer <i>Initialize 'pNSA' at the point of declaration</i>	CppParser/testsuite/src/CppParserTest.cpp	4
MISRA-007	WARNING	Variable 'pNSB' of type 'int *' declared without an initializer <i>Initialize 'pNSB' at the point of declaration</i>	CppParser/testsuite/src/CppParserTest.cpp	4
MISRA-007	WARNING	Variable 'missing' of type 'int' declared without an initializer <i>Initialize 'missing' at the point of declaration</i>	Crypto/src/EVPCipherImpl.cpp, Crypto/src/RSACipherImpl.cpp	4
MISRA-007	WARNING	Variable 'strPub' of type 'std::ostringstream' declared without an initializer <i>Initialize 'strPub' at the point of declaration</i>	Crypto/testsuite/src/EVPTest.cpp	4
MISRA-007	WARNING	Variable 'strPriv' of type 'std::ostringstream' declared without an initializer <i>Initialize 'strPriv' at the point of declaration</i>	Crypto/testsuite/src/EVPTest.cpp	4

Rule	Severity	Description	Files Affected	Occurrences
MISRA-007	WARNING	Variable 'strPub2' of type 'std::ostringstream' declared without an initializer <i>Initialize 'strPub2' at the point of declaration</i>	Crypto/testsuite/src/EVPTest.cpp	4
MISRA-007	WARNING	Variable 'key3' of type 'int' declared without an initializer <i>Initialize 'key3' at the point of declaration</i>	Crypto/testsuite/src/EVPTest.cpp	4
MISRA-007	WARNING	Variable 'strPub3' of type 'std::ostringstream' declared without an initializer <i>Initialize 'strPub3' at the point of declaration</i>	Crypto/testsuite/src/EVPTest.cpp	4

5. Knowledge Silo Alerts

Files where only one contributor has committed in the last 12 months. These represent bus-factor risk.

No knowledge silos detected.

6. Bug Prediction — Top 10 Highest-Risk Files

XGBoost model bug probability. Files above 0.7 should be prioritised for review.

#	File	Bug Probability	Risk Level	Top Factor
1	ActiveRecord/ testsuite/src/ ActiveRecordTestSuite.cpp	97.9%	CRITICAL	misra_findings
2	ActiveRecord/ testsuite/src/ WinDriver.cpp	97.9%	CRITICAL	misra_findings
3	CppParser/src/ Utility.cpp	97.9%	CRITICAL	misra_findings
4	CppParser/ testsuite/src/ AttributesTestSuite.cpp	97.9%	CRITICAL	misra_findings
5	CppParser/ testsuite/src/ CppParserTestSuite.cpp	97.9%	CRITICAL	misra_findings
6	CppParser/ testsuite/src/ WinDriver.cpp	97.9%	CRITICAL	misra_findings
7	Crypto/testsuite/ src/Driver.cpp	97.9%	CRITICAL	misra_findings
8	Crypto/testsuite/ src/ECTest.cpp	97.9%	CRITICAL	misra_findings
9	Data/SQLParser/ src/parser/ bison_parser.cpp	97.9%	CRITICAL	misra_findings
10	ActiveRecord/ Compiler/src/ CodeGenerator.cpp	97.2%	CRITICAL	misra_findings

7. Refactoring Roadmap

Prioritised by impact score. Complete high-impact, low-effort items first.

Priority	File	Action	Category	Est. Hours	Impact
1	ActiveRecord/ Compiler/src/ ImplGenerator. cpp	Reduce cyclomatic complexity by extracting methods and simplifying control flow	COMPLEXITY	3.0h	16.0
2	ActiveRecord/ Compiler/src/ ImplGenerator. cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	2.0h	16.0
3	Benchmark/src/ BenchmarkApp.c pp	Reduce cyclomatic complexity by extracting methods and simplifying control flow	COMPLEXITY	9.0h	16.0
4	Benchmark/src/ BenchmarkApp.c pp	Address MISRA C++ compliance violations to meet safety standards	MISRA	26.0h	16.0
5	ActiveRecord/ include/Poco/ ActiveRecord/ Query.h	Fix memory safety issues: replace raw pointers with smart pointers and add bounds checks	MEMORY SAFETY	8.0h	16.0
6	ActiveRecord/ include/Poco/ ActiveRecord/ Query.h	Address MISRA C++ compliance violations to meet safety standards	MISRA	6.0h	16.0
7	ActiveRecord/ Compiler/src/ CodeGenerator. cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	12.0h	16.0
8	Benchmark/src/ LoggerBench.cp p	Address MISRA C++ compliance violations to meet safety standards	MISRA	36.0h	16.0

Priority	File	Action	Category	Est. Hours	Impact
9	ActiveRecord/ Compiler/src/ Parser.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	10.0h	16.0
10	ActiveRecord/ Compiler/src/ Compiler.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	8.0h	16.0
11	ActiveRecord/ Compiler/src/ HeaderGenerator.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	6.0h	16.0
12	ActiveRecord/ testsuite/src/ ActiveRecordTest.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	58.0h	16.0
13	ActiveRecord/ testsuite/src/ ActiveRecordTestSuite.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	2.0h	16.0
14	ActiveRecord/ testsuite/src/ WinDriver.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	2.0h	16.0
15	ApacheConnector/samples/ FormServer/ src/ FormServer.cpp	Fix memory safety issues: replace raw pointers with smart pointers and add bounds checks	MEMORY SAFETY	4.0h	16.0
16	ApacheConnector/samples/ FormServer/ src/ FormServer.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	14.0h	16.0
17	ApacheConnector/samples/ TimeServer/ src/ TimeServer.cpp	Fix memory safety issues: replace raw pointers with smart pointers and add bounds checks	MEMORY SAFETY	4.0h	16.0
18	ApacheConnector/samples/ TimeServer/ src/ TimeServer.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	2.0h	16.0

Priority	File	Action	Category	Est. Hours	Impact
19	ApacheConnector/src/ ApacheConnector.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	12.0h	16.0
20	Data/SQLParser/src/ SQLParser.cpp	Address MISRA C++ compliance violations to meet safety standards	MISRA	4.0h	12.0

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